1. Identity

Document	STAR-PKG-STAR-SIGMA-EN · v2.1 · 2026-11-01
Package	star-sigma
Version	0.1.0
Capability	star.statistics.sigma-coverage
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sigma

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_sigma-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_sigma-0.1.0-py3-none-any.whl	267e6dd0657df13d69d07e6e21a4e27871b4dab50b4785471c1baec0ca27b7f9
star_sigma.py	84ed2154cc1d8d7a7df7f4075731af5eff78ed90034aeb5906540a7550a7e748
star_sigma-0.1.0-source.zip	f871c61c6ae254eee5dc4307f8e6a7c8fdff410ee806ed2a7fdceb267cafc4ad

5. What it does, and its limits

Coverage, tail probability and inverse of n sigma for a Gaussian error in one, two and three dimensions, and the normal distribution function and quantile; within 1.9e-13 relative of mpmath at 40 digits and 2.4e-13 of SciPy on 600 cases, tails included; reproduces the published normal and chi-square tables.

A Gaussian error with a known covariance: real orbit-determination errors are often not Gaussian and the covariance is often optimistic, so these probabilities are those of the model, not of the truth. No collision probability: that needs the geometry of the encounter. Dimensions 1, 2 and 3 only. The 1.9e-13 relative error in the far tails comes from the platform's erfc. A coverage given as 1 − 1e-17 cannot be distinguished from 1: use the tail.

- `normal_cdf`, `normal_sf`, `normal_quantile` for the standard normal distribution, accurate in both tails. Reproduce the published table values Φ(1), Φ(1.959964), Φ(2.575829).
- `sigma_coverage(n, dims)` and `sigma_tail(n, dims)` for dims 1, 2, 3 (the chi-square law of the squared Mahalanobis distance), each computed directly so that neither loses digits; `sigma_for_coverage(p, dims)` is the inverse. Reproduce the 68-95-99.7 rule and the published chi-square percentage points for 1, 2 and 3 degrees of freedom.
- Measured on 600 cases (z to ±37, n from 1e-4 to 12, probabilities from 1e-12 to 1 − 1e-12), as RELATIVE differences: distribution functions and tails within 1.9e-13 of mpmath at 40 digits and 2.4e-13 of SciPy; coverage within 8e-16 of mpmath; the inverses reproduce the requested tail probability within 1.3e-14.
- Every function returns a finite float or raises `ValueError`: non-numeric, boolean, NaN or infinite input, z beyond ±38, n outside [0, 38], a probability outside the open or half-open interval stated, dims not 1, 2 or 3.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_sigma_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-sigma}: What "n sigma" means in one, two and three dimensions},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sigma/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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